

Experiment 11 – simple calculator program

CSA1717 – Artificial Intelligence

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Aim:

To write a Python program that implements a simple calculator to perform basic arithmetic operations such as addition, subtraction, multiplication, and division.

Algorithm:

1. Start the program.
2. Display the menu with the following options:
 - a. Addition
 - b. Subtraction
 - c. Multiplication
 - d. Division
 - e. Exit
3. Read the user's choice.
4. If the choice is 5, terminate the program.
5. If the choice is not between 1–4, display an error message and ask again.
6. Read two numbers from the user.
7. Based on the choice:
 - If choice = 1 → add the numbers
 - If choice = 2 → subtract the numbers
 - If choice = 3 → multiply the numbers
 - If choice = 4 → divide the numbers (check division by zero)
8. Display the result.
9. Repeat the process until the user chooses to exit.
10. Stop the program.

Source Code:

Simple Calculator Program

while True:

print("\n--- Simple Calculator ---")

print("1. Addition")

print("2. Subtraction")

print("3. Multiplication")

print("4. Division")

print("5. Exit")

choice = int(input("Enter your choice (1-5): "))

if choice == 5:

print("Exiting the calculator. Goodbye!")

break

if choice not in [1, 2, 3, 4]:

print("Invalid choice! Please try again.")

continue

num1 = float(input("Enter first number: "))

num2 = float(input("Enter second number: "))

if choice == 1:

result = num1 + num2

print("Result:", result)

elif choice == 2:

```
result = num1 - num2  
print("Result:", result)
```

```
elif choice == 3:
```

```
result = num1 * num2  
print("Result:", result)
```

```
elif choice == 4:
```

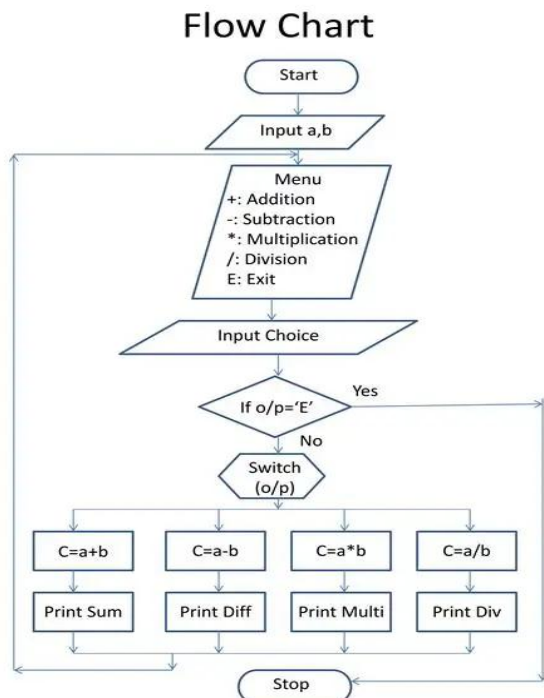
```
if num2 == 0:
```

```
    print("Error! Division by zero is not allowed.")
```

```
else:
```

```
    result = num1 / num2  
    print("Result:", result)
```

Flowchart:



Output:

EXP 11_output_AI.py - C:\Users\hemal\OneDrive\Documents\SIMATS\AI\AI_PY\EXP 11_outp

File Edit Format Run Options Window Help

```
--- Simple Calculator ---
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice (1-5): 1
Enter first number: 23
Enter second number: 34
Result: 57.0

--- Simple Calculator ---
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice (1-5): 2
Enter first number: 23
Enter second number: 76
Result: -53.0

--- Simple Calculator ---
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice (1-5): 3
Enter first number: 3
Enter second number: 5
Result: 15.0

--- Simple Calculator ---
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice (1-5): 4
Enter first number: 2
Enter second number: 6
Result: 0.3333333333333333

--- Simple Calculator ---
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice (1-5): 5
Exiting the calculator. Goodbye!
```

Result:

The program successfully implements a simple calculator that performs addition, subtraction, multiplication, and division based on the user's choice. It also handles invalid choices and division by zero.